# A kinetic-impact deflection model



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Estimate a velocity change and explain the value of lead time.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 3, Day 8; 3 course days later.*
0 m/s; equal opposite momenta cancel, and the combined mass is 6 kg.

**02. Explain a worked example (worked)**
Initial momentum 60 kg·m/s; joined mass 12 kg moves at 5 m/s.

**03. Look and explain (model)**
Longer time; shift grows as$\Delta v$ times time.

**04. Build the solution (practice)**
1 m and 10 m.

**05. Use the idea (reason)**
Ejected material carries momentum, target properties matter and orbital geometry changes the response. Simplified model omits these.

**06. Choose a method (mixed)** *Review: Unit 2, Day 15; 12 course days later.*
Need orbital geometry/positions and phase or time reference. A period alone does not show simultaneous nearby positions; uncertainties matter.

**07. Exit check (exit)**
Twice the shift in straight-line model; ignores gravity/orbital geometry and target uncertainties.

**08. STRETCH: Apply it another way (stretch)**
60/22≈2.73 m/s; same total momentum spread over greater final mass.

**Teaching note:** Science reference: NASA, “NASA’s DART Data Validates Kinetic Impact as Planetary Defense Method.” All numerical impact cases here are hypothetical.

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**01. Bring back an earlier idea (retrieve)** *Review: Unit 3, Day 8; 3 course days later.*
$v_f$=5 m/s. Initial K=375 J; final K=37.5 J; fraction 0.10 or 10%.

**02. Explain a worked example (worked)**
Initial momentum 60 kg·m/s; joined mass 12 kg moves at 5 m/s.

**03. Represent and interpret (model)**
Longer time; shift grows as$\Delta v$ times time.

**04. Solve and justify (practice)**
Momentum $3.6\times10^{6}$ kg·m/s;$\Delta v$ $7.2\times10^{-4}$ m/s; shift 720 m. Values are a classroom model, not the measured DART result.

**05. Analyze the reasoning (reason)**
Ejected material carries momentum, target properties matter and orbital geometry changes the response. Simplified model omits these.

**06. Choose a method (mixed)** *Review: Unit 2, Day 15; 12 course days later.*
Overlap days 18-20. Timing is compatible, but spatial geometry, miss distance and uncertainties are still needed.

**07. Exit check (exit)**
$\Delta v$=.005 m/s; impulse $5\times10^{6}$ N·s under the simplified transverse model.

**08. STRETCH: Apply it another way (stretch)**
20 N·s; 200 N; subsequent offset 2000 m. This simplified drift ignores gravity and orbital geometry.

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